Six-week data health report

Ridgeline Schools Weather Network · 8 stations

Prepared for the network coordinator
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Six of the eight stations' data needs no changes. The other two have problems that are easy to work around once you know where they are. Full detail, including the exact dates and the evidence for each call, is in `flags.csv` in the project repository.

1. Which data is good enough for the spring climate summary?

STATION	STATUS	WHAT TO DO
S01 Saddle Ridge Hut	Use as is	Nothing found wrong across six weeks.
S02 North Mesa School	Use pressure with caution	Pressure jumps ~7 hPa in under an hour on **Mar 21, ~17:00 UTC** (11am local) and never comes back — matches the volunteer's note about swapping the pressure board that morning without calibrating it. Temperature, humidity, wind, rain and sunlight are unaffected. Drop pressure after Mar 21 from the summary until it is recalibrated against a reference.
S03 Valley Library	Exclude temperature	Temperature reads an impossible −40 °C or 85 °C at four separate moments (classic sensor fault codes), and is frozen dead flat at exactly 6.5 °C for 80 hours straight, **Mar 18–21**. All other channels at S03 look fine.
S04 Foothills Community Garden	Use as is	Nothing found wrong. (The repainted fence post by the sensor, noted Mar 15, shows no measurable effect on sunlight readings when checked against a nearby station.)
S05 Eastside Rec Centre	Exclude humidity window; fix location	Humidity sticks at 100–104% for over two days, **Mar 7 13:30 to Mar 11** (the saturated sensor behind the "why does it say 100% all week" question from the kids). Separately, this station's own GPS consistently reports a location 20.5 km from the coordinates on file — almost certainly a typo in the install sheet, not a station that moved. Worth a one-line fix in `stations.csv` before anyone maps this network.
S06 South Hills Fire Station	Fix timestamps after Mar 24	After the battery swap on **Mar 24**, the logger clock reset: six hours of readings (10:10–16:00) got written under duplicate, colliding timestamps and should be dropped. From that point on, every GPS fix shows the logger running exactly 6 hours behind true UTC — consistent with the clock falling back to local Mountain Daylight Time instead of UTC and never being corrected. The weather readings themselves look fine; only the clock is wrong. Shift timestamps from Mar 24 16:00 onward forward by 6 hours before using this station in any time-aligned comparison.
S07 Riverbend Farm	Wind not comparable after Mar 26	Average wind speed roughly halves right when the anemometer was swapped for a different brand on **Mar 26** (about 10 km/h before, 4 km/h after), while neighbouring stations show no such drop that week. Treat wind speed before and after Mar 26 as two different instruments until the new unit is checked against a reference.
S08 Plains Grange Hall	Exclude two windows	No data at all for 35 hours, **Mar 16 14:00 to Mar 18 01:30** (the SIM data outage). Separately, **Apr 5** is logged twice with identical values — keep the first copy, drop the second.

2. Where should the technician go next week?

Send them to S03 - Valley Library

1. **Check first:** the temperature sensor's cable and connector at the logger. A connection that is intermittently open produces exactly what the data shows: brief readings that snap to a fixed fault value (−40 or 85 °C), recurring roughly weekly, plus one stretch where it stuck at a single value for over three days instead of faulting cleanly.
2. If the connector and cable look fine, swap the temperature sensor itself — the fault pattern only affects temperature, not the other channels on the same logger, which points at the sensor or its lead rather than the logger board.
3. While there, do a quick visual check of the radiation shield (spider webs were noted here on a routine visit last month) since it is a five-minute job on the same trip.

Why not S02 (the next most obvious candidate)? S02's pressure problem is a known, fully-diagnosed calibration issue — the board was already swapped, it just needs a reference instrument and ten minutes to calibrate. It doesn't cost any data in the meantime beyond pressure. S03's fault is still active, costs an entire channel for stretches of days at a time, keeps recurring, and the cause has to be found on site. It is the station actually losing data every week, so it gets the one hourthis week.